

Programme: B E – Information Science & Engineering
Internal Assessment – I

TERM : 10/09/2025 to 31/12/2025	COURSE NAME: Software Engineering
DATE : 28/10/2025 TIME: 9.30 AM to 10.30 AM	COURSE CODE : 23IS51
MAX MARKS: 30	PORTIONS : L1 - L19



Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Define Software Engineering. How is it more than "just programming"? Consider a client has a vague idea of their requirements and expects them to change frequently. Which process model would you recommend and why?	L3	CO1	8
b	Differentiate between Functional and Non-Functional Requirements (NFRs). Provide two examples of each for an e-commerce website.	L3	CO2	7
2.a	With diagram, explain in detail about the waterfall process model for software development.	L2	CO1	8
b	What are the common challenges faced during requirements elicitation, and how can a software engineer mitigate them?	L2	CO2	7
3.a	Describe and elaborate the four distinct phases of the Rational Unified Process (RUP), with a neat diagram.	L2	CO1	6
b	Create a simple Use Case Diagram for a "Library Management System" involving actors like Member and Librarian, with at least three key use cases.	L3	CO2	5
c	list & briefly explain requirement validation process.	L2	CO2	4

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create

Programme: B E – Information Science & Engineering
Internal Assessment – II

TERM : 10/09/2025 to 31/12/2025	COURSE NAME: Software Engineering
DATE : 23/12/2025 TIME: 11.00AM to 12.00	COURSE CODE : 23IS51
MAX MARKS: 30	PORTIONS : L20 - L40



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Instructions to Candidates: **Answer any TWO full questions.**

Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Illustrate advantages and disadvantages of the repository and layered architectural design models.	L2	CO3	4
b	Demonstrate equivalence class test cases for the NextDate function..	L3	CO4	6
c	Explain the factors effecting software pricing.	L2	CO5	5
2.a	With a neat diagram, explain the architecture of a language processing system.	L2	CO3	4
b	Using boundary value testing, write the test cases of an improved version of triangle problem that takes 3 integers as sides of the triangle and finds whether the triangle is scalene, Isosceles and Equilateral.	L3	CO4	6
c	Explain why the process of project planning is iterative and why a plan must be continually reviewed during a software project.	L2	CO5	5
3.a	Briefly Explain any two software design patterns with suitable diagrams.	L2	CO3	4
b	Define Equivalence class testing. Write weak robust equivalence class test cases for Commission Problem.	L3	CO4	5
c	Discuss strategies used for risk management and managing people at a workplace in an organization.	L2	CO5	6

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create